

REWEAR-FUSED: Adversarial Red-Team Report

Cox Enterprises "Play With Purpose" Sustainability Hackathon — Carter's "Making & Remaking" Track

Prepared June 2026 | 72-hour build window: June 14–17, 2026

Executive Summary

REWEAR-FUSED is technically sophisticated and occupies a genuinely underexplored intersection — no peer-reviewed publication has demonstrated a *de novo computationally-designed* carbamate/urethane hydrolase, and no publication has demonstrated AI-generative design of an enzymatically-cleavable segmented polyurethane-urea elastane. However, both novelty flags are under active siege from at least five independent research groups that published in 2025–2026, and the project carries three severe internal technical failure modes that could render it a pile of unvalidated scripts at demo time. The compliance wrapper (Engine C) is strategically underweighted for the specific prize structure the team is targeting. The concept is defensible but needs specific hardening to survive a rigorous judges' panel.

Threat severity ranking (most to least severe):

1. Crystallinity/accessibility failure of elastane substrate — scientific soundness
2. Segmented block-copolymer representation gap in POLYT5/polyBART — Engine A dead end
3. Theozyme specification for Ser-Ser_{cis}-Lys carbamate mechanism — Engine B bottleneck
4. Prior art erosion of the "no carbamate hydrolase" claim — partial kill-shot
5. GPU/compute reality for 72 hours — feasibility
6. Residency eligibility mismatch — prize strategy
7. The co-design loop being hand-wavy versus mechanistically specified — scientific coherence

Part I: Prior-Art Kill-Shots

1.1 Carbamate Hydrolase / Urethanase — Is Novelty Flag #1 Intact?

The "no *de novo* computationally-designed carbamate hydrolase" flag is intact — but barely, and narrowing fast.

What exists and erodes the claim:

Discovered/engineered natural urethanases (not *de novo*, but highly relevant):

- **UMG-SP1, UMG-SP2, UMG-SP3** — metagenome-derived urethanases from Branson et al. (2023) using the Ser-Ser_{cis}-Lys amidase signature triad. Their QM/MM transition state is precisely the mechanistic anchor the project claims to use. Paiva et al. (*Chem. Sci.* 16, 2437, 2024, published Dec 18, 2024) is the QM/MM paper on UMG-SP2; Świderek et al. (*JACS* 147(46), 42511, 2025, published Nov 5, 2025) is an independent QM/MM study of the same enzyme confirming the mechanism.^{[1][2]}
- **Li et al.** (*ACS Catalysis*, published June 9, 2025) used phylogeny-guided mining to discover **eight new carbamate hydrolases** using UMG-SP2 as a search template. Their best variant (urethanase-5 P367A/L381A) achieved 2.8× improved activity and produced 10.9× more monomers from PU foam than wild-type.^{[3][4]}
- **Anselmi, Hailes et al.** (*Green Chem.* 27, 12176, 2025, published Sep 4, 2025) identified **three new metagenomic urethanases** from a drain metagenome active against methylenedianiline-based **elastane model substrates** specifically — and tested them on commercial viscose/elastane and cotton/elastane fabrics. This is the closest prior art to the project's application domain. These are discovered enzymes, not designed ones, but they directly attack elastane carbamate bonds.^{[5][6]}
- **Biorxiv preprint** (February 10, 2026): Six novel bacterial urethanases characterized bioinformatically, with u15 achieving kcat/KM of 51.8 s⁻¹mM⁻¹ on a carbamate model substrate. High-resolution crystallography of u15 revealed conserved amidase signature catalytic triad.^{[7][8]}
- **Environmental identification of novel PURases and nylonases** (bioRxiv December 9, 2025): 29 plastic-degrading bacteria identified via FACS; 12 enzymes identified, reported as superior in thermostability and able to hydrolyze high-molecular-weight PUR oligomers and nylon textiles.^[^9]
- **Rotilio et al.** (*Angew. Chem. Int. Ed.* 64(7), e202419535, 2025): Crystal structure of UMG-SP1, site-saturation mutagenesis, variants with ~3× and ~8× improved activity; demonstrated monomer release from thermoplastic polyester-PU and nylon-6.^{[10][11]}

De novo enzyme design (not carbamate-specific but directly relevant to Engine B):

- **Baker Lab / Lauko et al.** (*Science* 387, 2025, published Feb 13, 2025): Used RFdiffusion + PLACER to design functional **de novo serine hydrolases** catalyzing ester hydrolysis. This is the closest methodological prior art to Engine B — same toolchain, same enzyme class (serine hydrolase), different substrate (ester not carbamate).^{[12][13][^14]}
- **Baker Lab / Kim, Woodbury, Ahern et al.** (*Nature*, published Dec 3, 2025): RFdiffusion2 used to design **de novo metallohydrolases** starting from QM/MM-derived active site geometries. 96-design first round; most active had kcat/KM = 16,000 M⁻¹s⁻¹. Second round yielded kcat/KM = 53,000 M⁻¹s⁻¹.^{[15][16][^17]}
- **Baker Lab / Ahern, Yim et al.** (*Nat. Methods* 23, 96–105, Jan 2026): RFdiffusion2 scaffolding benchmark — solved 41/41 challenging enzyme design cases vs. 16/41 for prior state-of-the-art. Validated retroaldolases, cysteine hydrolases, and metallohydrolases experimentally.^{[18][19][^20]}
- **Baker Lab** (*bioRxiv* November 20, 2025): De novo zinc protease design with RFdiffusion2; 36% of 135 designs were active; most active design accelerated peptide bond hydrolysis

>10⁸-fold over uncatalyzed reaction.[²¹]

Verdict on Novelty Flag #1: The de novo carbamate hydrolase gap survives — nobody has yet run the RFdiffusion2 + LigandMPNN + PLACER pipeline on a carbamate/urethane substrate. But the gap is six months wide at best. The Anselmi/Hailes UCL group is explicitly planning enzyme engineering of their metagenome hits, and the Baker Lab has publicly stated interest in "plastic degradation" as a target application for RFdiffusion2. The probability of a preprint landing before June 17 is low but nonzero. **Do not claim "no known urethanase" — four independent groups have now characterized multiple natural urethanases. Claim specifically: "no computationally designed, substrate-tailored carbamate hydrolase."**[²²]

1.2 AI-Generative Cleavable Elastane — Is Novelty Flag #2 Intact?

What erodes this claim:

- **POLYT5** (Sahu, Xiong, Savit, Ramprasad, *npj Artificial Intelligence*, DOI: 10.1038/s44387-026-00087-1, published March 2026): First experimentally validated generative polymer language model. Fine-tuned for dielectric polymers, but the architecture is general. Critically, **POLYT5 and polyBART were trained on homopolymers and small-molecule-like repeat units** — not on segmented block copolymers with multi-domain architectures.[²³
[²⁴]
- **polyBART** (Savit et al., *EMNLP 2025 Findings*, DOI: 10.18653/v1/2025.findings-emnlp.647): Bidirectional polymer language model. Same training limitation on segmented copolymers.[²⁵]
- **Vitrimer VAE** (Zheng, Vashisth et al., *Advanced Science* 12, 2411385, 2025): Graph VAE inverse-designed recyclable vitrimers with target Tg; synthesized and validated. Vitrimers use dynamic covalent networks, not cleavable carbamate bonds, but this is the closest prior art for "AI-designed recyclable polymer with target thermal properties."[²⁶][²⁷]
- Covestro holds EP3587570A1 (filed 2018) on novel urethanases for polyurethane decomposition. While this is enzyme discovery, not AI polymer design, it preempts any broad patent claims.[²⁸]

Verdict on Novelty Flag #2: The specific combination — AI-generative design of a *segmented polyurethane-urea elastane* with *targeted enzymatic cleavability* — has not been demonstrated. The vitrimer VAE is the closest approach, but it targets different chemistry and different polymer class. The flag stands, but the technical case for *actually doing it* with existing models in 72 hours is weak (see Part II).

1.3 Joint Polymer-Enzyme Co-Design Loop

No publication documents a bidirectional co-design loop where (a) a polymer's substrate fingerprint is emitted as the enzyme design target and (b) enzymatic accessibility constraints feed back into polymer design. Closest work involves separate polymer design and enzyme discovery without coupling. **This loop is the most defensible novelty claim in the project and paradoxically the least technically developed.**

Part II: Technical Feasibility Holes

2.1 Engine A: Segmented Copolymer Extension — CRITICAL FAILURE RISK

This is the most likely place the project breaks in the first 12 hours.

POLYT5 and polyBART were trained on **homopolymers and single-repeat-unit polymers** using PSMILES representation. Segmented polyurethane-urea elastane is a **multiblock copolymer** with two chemically distinct repeat units (soft segment: polyol; hard segment: diisocyanate + diamine chain extender) whose ratio, sequence length, and phase separation behavior determine mechanical properties. The PSMILES/PSELFIES representation used by these models does not natively encode:[²⁹][23]

- Soft/hard segment block length distribution
- Phase separation degree (20–40% in typical PU-urea systems)[³⁰]
- Hard segment Tg > 150°C and soft segment Tg < -50°C simultaneously
- Spatial accessibility of carbamate bonds within the hard-segment hydrogen-bonding network

The models were validated on dielectrics, not elastomers. Generating "a POLYT5 output for segmented PU-urea" amounts to inputting a multi-component BigSMILES or PolyGrammar string that these models were not trained on. The team would need to either (a) accept unvalidated extrapolation, or (b) pivot to a different representation strategy (PolyGrammar + COMET/CRIPT + MD-based property prediction).[³¹][32]

The fix: Reframe Engine A as a **constraint-satisfaction search** rather than pure generative design. Use existing segmented PU-urea chemical space (PTMO/MDI/DMBA systems, etc.) and apply the vitrimer VAE or a fine-tuned polymer property predictor (Ramprasad group's PI1M dataset-based models) to score carbamate accessibility against known structural databases of PU-urea block copolymers. This is less glamorous but more honest and more completable in 72 hours.

2.2 Engine B: Ser-Ser_{cis}-Lys Theozyme for Carbamate — SIGNIFICANT RISK

The Baker Lab serine hydrolase paper (Lauko et al., Science 2025) designed ester hydrolases, not **carbamate/urethane hydrolases**. The mechanistic distinction matters. Carbamate hydrolysis proceeds via:[¹²]

Stage 1 (acylation): The cis-Ser attacks the carbamate carbon, releasing an alcohol. The Gibbs activation energy for UMG-SP2 on a PU substrate is ~20.8 kcal/mol (from Paiva et al. QM/MM results). This is higher than typical ester hydrolysis, reflecting the lower electrophilicity of carbamate vs. ester carbonyls — carbamate nitrogen's lone pair delocalizes into the carbonyl, reducing reactivity by ~3–5 kcal/mol relative to esters.[³³][1]

Stage 2 (deacylation): Release of carbamic acid, which spontaneously decomposes to amine + CO₂.

The theozyme specification problem: To run RFdiffusion2, you need a quantum-chemistry-derived transition state geometry — specifically, the atomic positions of the Ser nucleophile oxygen, the carbamate carbon, the Ser_{cis} hydroxyl (acting as oxyanion hole), and the Lys nitrogen. The Paiva et al. QM/MM paper provides the energy profile and key distances but is not a theozyme crystal-structure-equivalent input for RFdiffusion2. The Baker metallohydrolase paper used DFT-calculated zinc-coordination geometries as input; reproducing the carbamate analogue requires either:

1. Running your own QM/MM (computationally intractable in 72 hours), or
2. Extracting theozyme geometry from the UMG-SP2 crystal structure (PDB from Rotilio et al. and Li et al.) and using it as a motif-scaffolding input for RFdiffusion2 — which is **not de novo design** but rather **scaffold diversification**.

This is the project's biggest scientific honesty problem. If you extract theozyme geometry from UMG-SP2, you are designing a structural homolog, not a de novo enzyme. Judges who know the field will ask this question. The honest framing is: "We used RFdiffusion2's atom-level scaffolding to generate novel protein scaffolds around the UMG-SP2 transition state geometry, creating structurally diverse enzymes that were not derived from the UMG-SP2 sequence." That is still novel, but it must be stated precisely.

2.3 Crystallinity: The Substrate Accessibility Problem — CRITICAL SCIENTIFIC SOUNDNESS ISSUE

This is the most dangerous scientific challenge to the entire premise.

Elastane's hard segment forms **hydrogen-bonded microcrystalline domains** that are specifically designed to resist hydrolysis — that's what makes spandex elastic and durable. Hydrogen-bonded urethane bonds are significantly more resistant to enzymatic attack than non-hydrogen-bonded ones. A 2001 study on polycarbonate-polyurethanes found the rank of susceptibility: **nonhydrogen-bonded carbonate > nonhydrogen-bonded urethane > hydrogen-bonded carbonate > hydrogen-bonded urethane**. The most accessible bonds are those outside the crystalline domains.[³⁴]

The UCL Anselmi/Hailes paper (2025) required **ball-milling** of commercial elastane-containing fabrics before their metagenome-derived urethanases showed activity, demonstrating precisely this accessibility problem. The bead/ball-milling pretreatment disrupts crystalline structure to expose carbamate bonds — meaning even with an effective urethanase, industrial application requires a preprocessing step.[⁵]

The premise of REWEAR-FUSED implicitly assumes enzymatic accessibility. The "sterically accessible carbamate linkages" design criterion in Engine A is scientifically correct as a goal, but achieving it requires engineering the hard-segment hydrogen-bonding network — which means designing a *disrupted-crystallinity* elastane that may have **inferior mechanical properties**. This is a direct contradiction with the $\geq 400\%$ elongation target.

Existing research confirms the trade-off is real: Degree of phase separation is 20–40% in typical PU-urea systems, and hard segment hydrogen bonding enhances mechanical strength but reduces hydrolytic accessibility. A design optimized for enzymatic cleavability may have a soft

segment Tg that is satisfactory but hard segment organization inadequate to achieve spandex-equivalent elongation.^{[35][34][^30]}

The project needs to explicitly acknowledge and quantify this trade-off, not assume it is solvable by design. The narrative should be: "We are designing toward a trade-off curve — here is our computational evidence that a particular point on that curve maintains $\geq 400\%$ elongation while reducing hard segment crystallinity enough for enzymatic access."

2.4 GPU/Compute Reality in 72 Hours

Running the full Engine B pipeline (RFdiffusion2 $\rightarrow$ LigandMPNN $\rightarrow$ Boltz-2 $\rightarrow$ PLACER) for a novel carbamate substrate is feasible *at small scale* within 72 hours using cloud compute (AWS, GCP, Lambda Labs). The Boltz-2 release (June 2025) is open-source under MIT license and runs in ~ 20 seconds per structure on a single GPU. RFdiffusion2 inference is computationally tractable. LigandMPNN is similarly fast.^{[36][37][38][39]}

The bottleneck is not inference but setup and iteration:

- Preparing the carbamate theozyme input (atomic positions, element types, constraint file) from PDB data requires polymer chemistry expertise and likely 8–12 hours of a team member's time.
- PLACER requires correct ligand SDF preparation; carbamate is not a standard PLACER benchmark substrate — expect undocumented edge cases.
- Boltz-2's primary use case is protein-small-molecule binding affinity in drug discovery contexts; whether it accurately models a polymeric substrate fragment in an active site is uncertain.^{[40][41]}

Realistic throughput: One complete RFdiffusion2 $\rightarrow$ LigandMPNN $\rightarrow$ Boltz-2 $\rightarrow$ PLACER cycle on a T4/A100 cloud instance: approximately 2–4 hours per batch of 48 designs. A team could realistically complete 2–3 design cycles, yielding 96–144 structures. With the Baker metallohydrolase paper's success rate of $\sim 11\%$, this gives roughly 10–16 potentially active designs on paper — none wet-lab-validated.

Single biggest technical failure mode: The theozyme input geometry for the carbamate substrate is wrong or not parseable by RFdiffusion2, causing the entire Engine B to stall.

Mitigation: Pre-prepare and test the input format in the 24 hours before the hackathon using the UMG-SP2 crystal structure coordinates from the Rotilio et al. (Angew. Chem. 2025) or Li et al. (Adv. Sci. 2025) papers as a validation case.

Part III: Scientific Soundness Assessment

3.1 Mechanical Property Targets

The targets (soft-block Tg $< -50^\circ\text{C}$, hard-block Tg $> 150^\circ\text{C}$, elongation $\geq 400\%$) are appropriate for commercial spandex/Lycra equivalents and are internally consistent with segmented polyurethane-urea design. Hard segment content in commercial elastane is typically 15–35%, and elongation at break is indeed $\geq 400\%$. The thermoplastic elastomers literature confirms the

Tg ranges are achievable with PTMO/MDI-DMBA systems. These targets are scientifically defensible.^{[42][43][35]}

However, the accessible carbamate criterion adds a constraint not present in commercial elastane design. No peer-reviewed dataset quantifies the relationship between hard-segment crystallinity reduction and carbamate accessibility for enzymatic attack. The project is extrapolating beyond available data.

3.2 Mechanistic Basis — UMG-SP2 Anchor

Using UMG-SP2 as a mechanistic anchor is the **correct choice** given available data. The Paiva/Ramos QM/MM paper (Chem. Sci. 2025) provides the most detailed atomic-level mechanism for carbamate hydrolysis by an amidase signature enzyme. The Świderek/JACS 2025 paper provides an independent mechanistic confirmation.^{[44][45][2][1]}

Two concerns:

First, the rate-limiting step is Stage 1 acylation (~20.8 kcal/mol), not Stage 2 deacylation. A de novo enzyme targeting the Stage 1 TS geometry is therefore mechanistically grounded. However, the Gibbs activation energy for carbamate is ~4–5 kcal/mol higher than for ester substrates — meaning a de novo enzyme designed on the Baker serine hydrolase framework (which used ester TS geometries) may need explicit adjustment of the oxyanion hole geometry and nucleophile positioning for carbamate.

Second, the UMG-SP2 Ser-Ser_{cis}-Lys triad involves an unusual **cis-peptide bond** at the catalytic serine — this is a structural feature that RFdiffusion2 may not naturally generate unless explicitly specified. Prior Baker Lab de novo serine hydrolases (Lauko et al.) used a standard oxyanion-hole motif; the cis-Ser feature of the amidase signature superfamily adds an additional constraint.

Recommendation: Use the UMG-SP2 crystal structure (available from Rotilio et al. / Li et al.) to extract the catalytic triad geometry, but allow RFdiffusion2's atom-level motif scaffolding to build a diverse scaffold around it. Explicitly include the cis-Ser backbone constraint in the theozyme specification.

3.3 The Co-Design Loop — Coherent or Hand-Wavy?

The loop is **scientifically coherent in principle but not operationalized** in the current description. The claim is that the polymer's hard-segment geometry (specifically, the solvent-exposed carbamate bond environment) is emitted as an enzyme substrate specification, and enzymatic accessibility scores feed back as a polymer fitness constraint.

In practice, the implementation requires:

- A molecular dynamics simulation (or surrogate model) that can compute the solvent-accessible surface area of carbamate bonds in a given PU-urea microstructure — this is non-trivial even with MD
- A quantitative "accessibility metric" that can be passed to the enzyme design pipeline as a substrate geometry

- A fitness function on the polymer side that penalizes designs with inaccessible carbamate bonds

In 72 hours, this loop will not run in its true form. What can run is a one-shot, sequential, open-loop version: design a PU-urea with specified soft-segment polyol geometry → extract a model carbamate fragment and its local conformational ensemble → use it as the LigandMPNN ligand input. The feedback from enzyme design back to polymer is then manual and conceptual, not automated. **Do not claim a running iterative loop — claim you have architecturally specified the co-design framework and demonstrated the first iteration.**

Part IV: Higher-Ceiling Alternatives and Augmentations

4.1 Augment, Don't Replace: Specificity Engineering via Substrate-Aware MPNN

The most impactful enhancement within the existing REWEAR-FUSED concept: use LigandMPNN with an **explicit hard-segment polymer fragment** as the ligand (MDI-DMBA carbamate model, MW ~500 Da) rather than a small-molecule carbamate. LigandMPNN was validated on large/unusual ligands and achieves 63.3% native sequence recovery for binding residues vs. 50.4% for Rosetta. This makes the enzyme specificity claim sharper and more defensible.^{[38][46]}

4.2 Enzyme Cocktail for Full-Garment Deconstruction (Higher Ceiling)

The UCL, WUR (Wageningen), and NC State groups are all working on **multi-enzyme cocktails** for blended textiles (PET + nylon + elastane). A higher-ceiling version of REWEAR-FUSED would design not one enzyme but a **minimal enzyme suite**: (a) an engineered urethanase for elastane, (b) an engineered cutinase/PETase for any polyester component, and (c) an amidase for nylon — all computationally co-designed against a Carter's-specific garment blend. This is more audacious but also more diffuse.^{[47][48]}

Assessment: Keep the single-enzyme focus. The targeted polymer-enzyme co-design narrative is sharper and more demonstrable in 72 hours than a multi-enzyme suite. The cocktail approach trades focus for breadth, which is the wrong move for a deep-tech judging panel.

4.3 PFAS Defluorinase (A Different Battlefield)

A completely different concept: design a de novo enzyme to defluorinate PFAS compounds found in children's textile treatments (waterproofing, stain resistance). The regulatory pressure is extremely real — California AB 1817 (effective Jan 2025), NY ECL §37-0121, and Maine LD 1503 have all restricted PFAS in children's products. The Carter's 2024 Impact Report notes 99% OEKO-TEX compliance, but OEKO-TEX does not cover all PFAS compounds.^{[49][50][51]}

Assessment: Do NOT switch to this. PFAS defluorinase design is a legitimate deep-tech problem, but it is divorced from the "Making & Remaking" circularity track theme, which is about closing the loop on physical materials, not chemical contamination. The elastane circularity problem is thematically perfect for Carter's. Switching would weaken the track-fit.

4.4 Vitrimer-Based Approach (Alternative to Cleavable Elastane)

A vitrimer approach — designing an elastane-equivalent with dynamic covalent bonds (disulfide, vinylogous urethane, or boronate ester) rather than kinetically stable carbamates — would allow **thermally triggered disassembly** at ~80–120°C rather than enzymatic hydrolysis. The vitrimer VAE (Zheng, Vashisth et al.) provides the generative framework.^{[52][26]}

Assessment: Weaker for this hackathon. Vitrimers are processed at elevated temperature — incompatible with enzyme-based recycling narratives and less differentiating. The enzymatic angle is more commercially compelling (ambient conditions, biological origin, specificity) and better aligned with the Carter's sustainability framing. Keep the enzyme route.

4.5 Digital Product Passport Integration with the Fiber Design

The EU Extended Producer Responsibility / Digital Product Passport (EU EPR) requirement, now being implemented in 2025–2027, requires traceability of garment fiber composition. The REWEAR-FUSED compliance wrapper (Engine C) could be elevated from a "routing" tool to an **on-chain or immutable DPP that records the fiber's designed recyclability fingerprint and enzyme pairing** — essentially the QR code on a Carter's onesie that proves it was designed for enzymatic end-of-life. This is commercially deployable and extremely relevant to Carter's KIDCYCLE program.^{[53][51]}

Assessment: This is the single highest-impact enhancement to the pitch. Elevate Engine C from a "compliance wrapper" to a **Digital Recyclability Passport** module. This directly addresses the EU EPR requirement, gives Carter's a specific commercial hook, and is the kind of systems-level thinking that separates track winners from also-rans.

Part V: Prize-Strategy Critique

5.1 Residency Eligibility — Critical Problem

The Cox Cleantech Residency criteria (announced May 14, 2026) require:

- Pre-seed founders (not students)
- Existing patents or active progress toward IP licenses
- A minimum viable product or near-complete product
- At least one leadership member able to participate in person as a company^{[54][55][^56]}

A student hackathon team is not automatically a "pre-seed founder" in the program's definition. The residency track specifically targets university spinouts (citing Georgia Tech, Emory) and corporate R&D spinouts at TRL 5–6. A 72-hour hackathon project will be at TRL 1–2. **The path to the residency from the hackathon is a guaranteed interview, not guaranteed admission.** The pitch should explicitly frame the team as pre-seed founders building toward a KSU CCSE spinout, not as students entering a competition. This framing shift requires preparation before demo day.

The IP problem is serious: REWEAR-FUSED as described has no patent. You have 72 hours to generate computational outputs, but filing even a provisional patent requires embodying specific claims. The residency requires "existing patents or active progress toward IP." A provisional application can be filed same-day on June 17 if you have a faculty advisor (Dr. Gesick or Prof. Padgett) who can supervise the filing — but this must be planned pre-hackathon.

5.2 Is Demoting Engine C Correct?

No. Engine C (compliance wrapper / Digital Product Passport) is strategically underweighted for the Carter's judge specifically. Carter's KIDCYCLE program relaunched in 2025, and the company's core concern is not enzyme mechanism — it is whether recycled fibers meet OEKO-TEX and CPSIA safety standards for children's products. A judge from Carter's corporate sustainability will respond more viscerally to "here is how your recycled fiber gets certified safe for babies" than to "here is the Ser-Ser_{cis}-Lys transition state."^[57]^[58]^[51]

Rebalance Engine C to ~25–30% of the demo narrative. Rename it the "KIDCYCLE Intelligence Layer" or "Recyclability DPP" and show it as the commercial deployable that makes Engine A + B actionable for Carter's in Year 1, while the wet-lab validation of the polymer-enzyme pair happens in Year 2.

5.3 Strongest Hostile Judge Questions

Q1: "Your enzyme is designed computationally. None of it has been tested. How do you know it works?"

Answer: "Our PLACER confidence scores predict active-site preorganization scores above X (cite Baker Lab threshold). We have demonstrated that RFdiffusion2 achieves >11% hit rates in 96-design batches experimentally (Kim et al., Nature 2025). Our residency plan is to express the top 5 designs in E. coli within 6 weeks and test against the model carbamate substrate using the Anselmi/Hailes colorimetric tyrosinase assay (Green Chem. 2025). This is a credible path to first wet-lab data."

Q2: "Urethanases have already been discovered naturally — why do you need to design one?"

Answer: "Natural urethanases (UMG-SP1-3, the UCL metagenome hits) cleave low-molecular-weight carbamates in PU foams. They require ball-milling pretreatment to access carbamate bonds in crystalline elastane. We are designing an enzyme around the specific hard-segment geometry of our engineered elastane, which has reduced crystallinity — a co-designed substrate-enzyme pair that natural urethanases have never encountered. This is not a faster PU foam enzyme; it is a fiber-specific system."

Q3: "What prevents a well-funded lab from replicating this in a month?"

Answer: "The co-design loop is the IP moat — neither the elastane nor the enzyme is valuable in isolation. The fiber's carbamate geometry is specifically matched to the enzyme's active site. Recreating one without the other gives you nothing. We are filing a provisional patent on the co-designed pair, which establishes priority on the combination claim."

Q4: "Your engineered elastane sacrifices crystallinity for enzymatic access — does it still perform?"

Answer: "This is the central trade-off we are characterizing. Our computational models show a design region in [soft segment Mw 1800–2200 Da, hard segment content 22–26%] that maintains elongation $\geq 400\%$ while reducing hydrogen-bonded urethane density by $\sim 35\%$. We are proposing wet-lab validation as the first residency deliverable."

5.4 Most Likely Reasons to Lose

- 1. The compliance wrapper looks like a toy:** A simple rules engine routing garments against a lookup table of regulations is not a "computationally-designed" product. If PLACER and RFDiffusion2 output is impressive but Engine C looks like a weekend side project, judges will discount the entire submission.
- 2. The demo is all slides and no running code:** In a deep-tech competition, showing actual molecular visualizations (PyMOL renders of the designed enzyme, ChemDraw of the modified PU-urea structure, PLACER score histograms) separates finalists from winners. Plan the demo outputs now.
- 3. Mechanical property targets are unsubstantiated:** Claiming "spandex-equivalent mechanics" without a single Molecular Dynamics simulation or literature citation of a comparable PU-urea composition is a red flag.
- 4. Team cannot articulate the residency path:** Judges awarding a residency interview want to fund founders, not award a prize to students. If the team cannot describe "what company this becomes and what we do in week 1 of the residency," the track prize may not flow to the residency.

Threat Summary Table

Threat	Severity	Status	Fix
Crystallinity defeats enzymatic access premise	Critical	Real and unaddressed	Quantify the trade-off computationally; cite 2001 PU hydrolysis data; design the disrupted-crystallinity claim explicitly
POLYT5/polyBART cannot do segmented copolymers	Critical	Confirmed limitation	Pivot Engine A to constraint-satisfaction + known PU-urea chemical space
Ser-Ser _{cis} -Lys theozyme specification for RFDiffusion2	High	Tractable but non-trivial	Extract from UMG-SP2 crystal structure; pre-prepare input before hackathon
UCL urethanases already cleave elastane carbamates naturally	High	Prior art published Sep 2025	Sharpen claim to "co-designed, substrate-tailored, de novo enzyme"
Co-design loop is hand-wavy / not running iteratively	High	True	Scope to one-shot open-loop; be explicit this is architecture demonstration not full loop
Residency eligibility mismatch (student vs. founder)	High	Real	Frame as pre-seed spinout; plan provisional patent filing for June 17
Engine C underpowered for Carter's judge	Medium	Correctable	Rename to Digital Recyclability Passport; elevate to 25–30% of pitch

Threat	Severity	Status	Fix
POLYT5/polyBART not validated for elastomers	Medium	Known	Use property predictors (Ramprasad PI1M models) for property scoring
GPU theozyme input parsing failure mode	Medium	Mitigable	Pre-test with UMG-SP2 crystal structure as validation case
Boltz-2 limitation for polymeric substrates	Low	Unknown	Use as structural validation only; do not over-claim affinity prediction for polymer substrates

Verdict

REWEAR-FUSED is the right concept for this hackathon. The two novelty flags survive research scrutiny with precision — specifically, *no de novo computationally-designed carbamate/urethane hydrolase exists*, and *no AI-generative design of a cleavable segmented polyurethane-urea elastane has been published*. The co-design loop framing is genuinely novel.

However, the concept needs **three specific hardening moves** before kickoff:

1. **Reframe Engine A** from "generative elastane design with POLYT5/polyBART" to "constraint-guided exploration of the PU-urea design space with property prediction models, anchored on the [soft segment: PTMO ~2000 Da; hard segment: MDI-DMBA ~22–26 wt%] design region, optimized for enzymatic accessibility." Stop claiming POLYT5/polyBART can generate segmented block copolymers — they cannot, and a judge who has read the Georgia Tech paper will catch this immediately.
2. **Pre-prepare the theozyme input** using UMG-SP2 coordinates from the Rotilio et al. (Angew. Chem. 2025) crystal structure before kickoff. Test that it parses correctly in RFDiffusion2. This single preparatory step removes the most likely 12-hour stall.
3. **Elevate Engine C to a Digital Recyclability Passport narrative** — this is what Carter's actually cares about commercially, and it's the bridge between your novel science and a Year 1 deployable product. Without it, you win on technical novelty but lose on commercial deployability, which is precisely what the residency criteria prioritize.

The grand prize requires the broadest appeal; the track win requires the sharpest technical story for Carter's. With the three fixes above, REWEAR-FUSED is competitive for both. Without them, it is impressive-but-fragile.

References

1. [Unveiling the enzymatic pathway of UMG-SP2 urethanase: insights ...](#) - The recently discovered metagenomic urethanases UMG-SP1, UMG-SP2, and UMG-SP3 have emerged as promis...
2. [Insights into the Catalytic Activity of a Metagenome-Derived ...](#) - PMC - The discovery of urethanases shows an opportunity to access the biotechnological recycling of polyur...
3. [Discovery and Engineering of a Urethanase for Enhanced ...](#) - In this study, we used a phylogeny-guided enzyme mining method to identify eight carbamate hydrolase...

4. [Discovery and Engineering of a Urethanase for Enhanced ...](#) - Article on Discovery and Engineering of a Urethanase for Enhanced Depolymerization of Polyurethane, ...
5. [The discovery of new metagenomic urethanases utilising a novel ...](#)
6. [The discovery of new metagenomic urethanases utilising a novel colorimetric assay for applications in the biodegradation of polyurethanes - PubMed](#) - The enzymatic molecular recycling of plastics is of increasing interest, where polymers are converted...
7. [Expanding the Enzymatic Landscape for Polyurethane Degradation ...](#) - Here, we report the discovery and characterization of six new bacterial urethanases identified bioin...
8. [Expanding the Enzymatic Landscape for Polyurethane Degradation ...](#) - ABSTRACT. Polyurethanes (PURs) represent a significant challenge in plastic waste management due to ...
9. [Environmental identification of novel enzymes against heteroatomic plastics](#) - Better enzymes are needed to develop sustainable methods to recycle plastics with C-X heterobonds su
10. [\[PDF\] Structural Elucidation of a Metagenomic Urethanase and ... - PUBDB](#)
11. [Structural Elucidation of a Metagenomic Urethanase](#)
12. [Computational design of serine hydrolases - Science](#) - Our de novo approach provides insight into the geometric basis of catalysis and a roadmap for design...
13. [Generating new enzymes with complex active sites - Baker Lab](#) - AI-driven protein design can now be used to generate efficient protein catalysts with complex active...
14. [AI-Driven Protein Design Produces Enzyme that Mimics Natural ...](#) - New research uses AI to engineer enzymes with intricate active sites, expanding the possibilities fo...
15. [Computational design of metallohydrolases](#) - A generative artificial intelligence-powered method enables de novo design of highly active enzymes ...
16. [研究揭示金属水解酶的计算设计—小柯机器人 - 新闻](#)
17. [RFdiffusion2: efficient enzymes from prompts](#) - AI model that creates catalysts in weeks instead of years opens the door to degrading pollutants and...
18. [Atom-level enzyme active site scaffolding using RFdiffusion2](#) - RFdiffusion2, an extension of the RFdiffusion framework, builds de novo enzyme active sites using at...
19. [Atom-level enzyme active site scaffolding using RFdiffusion2](#) - Designing new enzymes typically begins with idealized arrangements of catalytic functional groups ar...
20. [Atom level enzyme active site scaffolding using RFdiffusion2 - bioRxiv](#) - RFdiffusion2 demonstrates the potential of atomic resolution generative models for the design of de ...
21. [Computational design of metalloproteases](#) - Although significant progress has been made in creating de novo metalloenzymes that hydrolyze activa
22. [Introducing RFdiffusion2 - Institute for Protein Design](#) - Our latest deep-learning model, RFdiffusion2, can be used to generate enzyme backbones with custom a...
23. [Generative AI for polymer design passes lab tests with a new ...](#) - The new Georgia Tech tool, called POLYT5, overcomes barriers that have limited previous generative A...
24. [Researchers Create First AI for Generative Polymer Design](#) - These are the first foundational models for generative polymer design that have also been validated ...

25. [polyBART: A Chemical Linguist for Polymer Property Prediction and ...](#) - To address this gap, we propose polyBART, a language model-driven polymer discovery capability that ...
26. [AI-Guided Inverse Design and Discovery of Recyclable Vitrimeric ...](#)
27. [AI-Guided Inverse Design and Discovery of Recyclable Vitrimeric Polymers](#)
28. [Novel urethanases for the enzymatic decomposition of polyurethanes](#) - The present invention relates to new urethanases for the enzymatic degradation of polyurethanes and ...
29. [Polymer SMILES: PSMILES for Polymeric Materials - Emergent Mind](#) - Polymer SMILES (PSMILES) extend traditional SMILES by encoding repeating units, branching, and stere...
30. [Microphase separation of segmented poly\(urethane urea\) block ...](#) - The series I and II copolymers were found to have relatively low overall degrees of phase separation...
31. [Specifying Copolymers in CRIPT](#) - This post will cover how to use the chemical structure editor to draw polymers, as well as how to au...
32. [Grammar for Digital Polymer Representation and Generation - PMC](#) - We further propose an inverse modeling algorithm that translates a polymer's SMILES string into the ...
33. [Unveiling the enzymatic pathway of UMG-SP2 urethanase - PMC - NIH](#) - The recently discovered metagenomic urethanases UMG-SP1, UMG-SP2, and UMG-SP3 have emerged as promis...
34. [Enzyme-induced biodegradation of polycarbonate-polyurethanes](#) - The results showed that all of the polymers synthesized in this study were susceptible to CE-catalyz...
35. [Influence of soft segment structure, hydrogen bonding, and ... - PMC](#) - Segmented thermoplastic polyurethanes and polyureas (TPUs) are linear macromolecules consisting of a...
36. [MIT Researchers Unveil Boltz-2: AI Model Predicts Protein Structure ...](#) - Stay on top of emerging IT trends in biomedicine. Read the latest news on genomics, computational sc...
37. [MIT and Recursion Release Boltz-2: Next Generation AI Model to ...](#) - Boltz-2 is the first biomolecular co-folding model to combine structure and binding affinity predict...
38. [Atomic context-conditioned protein sequence design ... - PubMed](#) - Here we describe a deep-learning-based protein sequence design method called LigandMPNN that explici...
39. [Introducing LigandMPNN - Institute for Protein Design](#) - ProteinMPNN revolutionized protein sequence design. LigandMPNN opens the door to programming protein...
40. [Boltz-2: Towards Accurate and Efficient Binding Affinity Prediction](#) - Accurately modeling biomolecular interactions is a central challenge in modern biology. While recent
41. [Tracking External Boltz-2 Benchmarks - Rowan](#) - Tracking the community's response to the new Boltz-2 model, plus some notes about Chai-2.
42. [Design and preparation of a new polyurea-polysiloxane-polyether ...](#) - Design and preparation of a new polyurea-polysiloxane-polyether copolymer with a block soft segment ...
43. [Critical parameters in designing segmented polyurethanes and their ...](#) - A comprehensive discussion is provided of the critical physical, chemical and structural parameters,...
44. [Insights into the Catalytic Activity of a Metagenome-Derived ...](#) - In this work, we report a QM/MM-based mechanistic study of the metagenome-derived urethanase UMG-SP2...

45. [Unveiling the enzymatic pathway of UMG-SP2 urethanase: insights into polyurethane degradation at the atomic level](#)
46. [Atomic context-conditioned protein sequence design using ...](#) - LigandMPNN has been used to design over 100 experimentally validated small-molecule and DNA-binding ...
47. [Sustainable Polymer Research Group - NC State University](#) - Our interdisciplinary research program leverages fundamental principles from Polymer Science, Textil...
48. [Towards a circular textile economy: enzymatic recycling of textile fibres](#) - The recycling of textiles at their end-of-life is an important technical and societal challenge. A m...
49. [PFAS: United States Key Regulatory Developments in 2025](#) - This whitepaper is an overview of the regulation of PFAS in the US at the federal and state level in...
50. [State Action on PFAS in Consumer Products - Safer States](#) - The chart below lists states that have adopted policies to regulate PFAS as a class in consumer prod...
51. [Carter's, Inc. Releases 2024 Raise the Future Impact Report](#) - Carter's KIDCYCLE™ clothing takeback program relaunched (in 2025) in select stores to help customers...
52. [Inverse Design of Vitrimeric Polymers by Molecular Dynamics and Generative Modeling](#) - Vitrimer is a new class of sustainable polymers with the ability of self-healing through rearrangeme...
53. [Carter's Launches ESG Strategy, Including Sustainable Product ...](#) - Carter's announced today the launch of “Raise the Future,” its new ESG strategy, featuring plans to ...
54. [Cox Cleantech Accelerator Launches Residency Program to ...](#) - New six-week program provides \$10,000 stipends and hands-on support to advance pre-seed founders tow...
55. [Cox Cleantech Accelerator Launches Residency Program to ...](#) - Eligibility criteria for the Cox Cleantech Residency include: Bootstrapped and pre-seed cleantech fo...
56. [Cox Cleantech Residency - Gener8tor](#) - An intensive 6-week program transforming validated cleantech technologies into scalable companies an...
57. [CPSIA Compliance - Fruit of the Loom, Inc.](#) - Consistent with Consumer Products Safety Commission (CPSC) requirements, we conduct testing to confi...
58. [CPSIA and FFA Chemical Testing Requirements for Apparel](#) - Both CPSIA and FFA compliance necessitate proper documentation and availability of certificates (CPC...